

Enterprise SIEM & Threat Logging Architecture	
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1. EXECUTIVE SUMMARY & CONTINUOUS MONITORING STRATEGY

Purpose: The purpose of this architecture document is to outline the deployment, configuration, and governance of the enterprise Security Information and Event Management (SIEM) environment. This ensures comprehensive visibility into endpoint activity, network traffic, and cloud authentication to support rapid Incident Response (IR) and compliance with NIST and ISO 27001 continuous monitoring requirements.

Core Objectives:

- **Centralized Visibility:** Aggregate telemetry from diverse operating systems (Windows Server, Red Hat Enterprise Linux, AlmaLinux) and network appliances (pfSense).
- **Proactive Threat Detection:** Utilize signature-based and anomaly-based detection rules to identify unauthorized access, privilege escalation, and lateral movement.
- **Compliance Auditing:** Maintain immutable, long-term retention of critical security logs to satisfy regulatory audit requirements.

2. WAZUH SIEM ARCHITECTURE & LOG INGESTION

The enterprise utilizes **Wazuh** as the primary SIEM and Extended Detection and Response (XDR) platform, deployed within a segmented, secure network boundary.

- **Endpoint Agent Deployment:** Wazuh agents are actively deployed across the hybrid infrastructure to capture system-level telemetry. This includes Windows Server 2022 Domain Controllers capturing Active Directory authentication events and Group Policy modifications, and Linux workloads (RHEL 10 & AlmaLinux) monitoring kernel-level events, SSH login attempts, and file integrity metrics.
- **Network Telemetry:** Syslog forwarding is configured on the **pfSense** perimeter firewall to ingest dropped packet logs, firewall rule modifications, and NAT translations into the SIEM indexer.

- **Log Normalization:** The Wazuh indexer automatically parses, normalizes, and enriches raw log data against MITRE ATT&CK framework tactics and techniques, ensuring security analysts have actionable context for every event.

3. FILE INTEGRITY MONITORING (FIM) & MITRE ATT&CK MAPPING

To detect unauthorized modifications and satisfy NIST/ISO compliance baselines, automated detection rules are enforced across endpoints:

- **File Integrity Monitoring (FIM):** Wazuh continuously monitors critical system directories (such as *C:\Windows\System32* on Windows and */etc/*, */usr/bin/* on Linux) in real time. Any unexpected modification, deletion, or permission change on critical system binaries immediately generates a high-severity alert.
- **MITRE ATT&CK Alignment:** Custom detection rules map log anomalies directly to standard adversary tactics. For Initial Access and Privilege Escalation, rules detect repeated failed authentication attempts followed by administrator privilege execution. For Defense Evasion, the system alerts on any manual attempt to clear security event logs or terminate the Wazuh agent service.

4. ALERT TRIAGE, ESCALATION, & AUDIT RETENTION

- **Severity-Based Alerting:** Telemetry is classified from Level 1 (informational) to Level 15 (critical breach indicator). Alerts reaching Level 10 and above trigger immediate notifications to the incident response team.
- **Audit Retention & Compliance:** Log data is stored in write-protected, indexed storage with automated retention policies configured for 365 days. This ensures full traceability for external regulatory audits and forensic breach investigations.